

Hrudayangam Mehta

Binghamton, NY | mehtahruday@gmail.com | +1 (607) 601-0523

github.com/Hrudayangam | linkedin.com/in/hrudaymehta | hrudaymehta.vercel.app

RESEARCH EXPERIENCE

Research Assistant

iDRAMA Lab, Binghamton University | May 2024 – Present

- Current research: preference optimization and AI alignment, including vision-language model (VLM) alignment.
- In progress: exploring VLM-based video annotation for computational social science on social media.
- Co-authored policy-guided LLM evaluation research (EMNLP 2025).
- Co-authored *From Woofs to Words* (AAAI 2026; equal contribution).
- Co-inventor of *AI-Driven Floor Plan Understanding for Indoor/Outdoor Navigation*. U.S. provisional patent application 63/884,317, filed September 18, 2025.

Graduate Research Assistant

AIR Group, Binghamton University | Aug 2023 – May 2024

- Integrated speech-to-text into a robotic guide-dog interface for spoken interaction.
- Related publication: *Vision language models can parse floor plan maps* (Robot Learning (ELSP), 2025; co-author, equal contribution).
- Evaluated GPT-4o and Claude 3.5 Sonnet for floor-plan navigation; analyzed failures across map size, task complexity, and label density.

SELECTED PUBLICATIONS

From Woofs to Words: Towards Intelligent Robotic Guide Dogs with Verbal Communication. **AAAI 2026**.

The paper integrates LLM dialogue, symbolic planning, and verbalization; its system achieved 94.8% task-specification accuracy on 77 simulated service requests.

Evaluating Large Language Models for Detecting Antisemitism. **EMNLP 2025**.

The paper introduces Guided-CoT prompting for policy-guided antisemitism detection, improving classification performance across eight open-source LLMs.

Vision language models can parse floor plan maps. **Robot Learning (ELSP), 2025**.

The paper develops a pipeline from floor-plan images and start-goal prompts to navigation actions, evaluating spatial reasoning across map sizes, task complexities, and label densities.

EDUCATION

PhD in Computer Science, Binghamton University, SUNY

Aug 2024 – Present | Adviser: Jeremy Blackburn; iDRAMA Lab

MS in Computer Science, Binghamton University, SUNY

Jan 2023 – May 2024 | Artificial Intelligence track; GPA: 3.83/4.00

BTech in Computer Science and Engineering, VIT Bhopal

Jul 2019 – May 2023 | Artificial Intelligence and Machine Learning

TECHNICAL SKILLS

Programming: Python, SQL, C, C++, JavaScript, TypeScript

Machine learning: LLM evaluation, vision-language models, classification, prompting and decoding experiments, error analysis, OpenCV

Data and software: PostgreSQL, MongoDB, Faktory, Flask, Git, Linux, AWS